

# Exponential Equations without using log functions

$$\Rightarrow 4^x = 32$$

$$4^x = 16(2)$$

$$4^3 = 64, 4^2 = 16$$

$$\text{so: } 2 < x < 3$$

$$\frac{64}{2} = 32 = \frac{4^3}{2} \Rightarrow 4^x = \frac{64}{2}$$

$$4^{\frac{1}{2}} = \sqrt{4} = 2 \Rightarrow 2(16) = 32$$

$$\text{then: } 4^2(4^{\frac{1}{2}}) = 16(2) = 32$$

$$\text{so } x = 4^{\frac{5}{2}} = \sqrt[5]{4} = 2^5 = 32$$

$$x = 2.5$$